

Module 15) Advance Python Programming

4. Reading and Writing Files

- **Reading from a file using read(), readline(), readlines().**

Python provides several methods to read the contents of a file once it is opened in **read mode ('r')** or a mode that allows reading ('r+', 'w+').

1. read() – Read Entire File as a Single String

- Reads the **entire content** of the file at once.
- Useful for small to moderately sized files.

Key Points:

- Returns the entire file content as one string.
- You can specify the number of characters to read: read(n).

2. readline() – Read One Line at a Time

- Reads the **next line** from the file.
- Useful for processing files line by line.

Key Points:

- Includes the newline character \n at the end of each line.
- Returns an empty string when the end of the file is reached.

3. readlines() – Read All Lines as a List

- Reads all lines and **returns them as a list of strings**.

- Each element in the list is one line (including the newline character).

Key Points:

- Suitable for reading the file into memory and then processing it line by line using loops.

• Writing to a file using write() and writelines()

Python provides the write() and writelines() methods to output data to files. These methods are used after opening a file in a **write-enabled mode** like 'w', 'a', 'r+', or 'w+'.

1. write() – Writing a Single String to a File

- Writes a **single string** to the file.
- Does **not automatically add a newline** (\n), so you must include it manually if needed.

Key Points:

- Overwrites content if the file is opened in 'w' or 'w+'.
- Appends if the file is opened in 'a'.

2. writelines() – Writing a List of Strings

- Writes **multiple strings** (from a list or iterable) to the file.
- **Does not add newlines** automatically – each string in the list must end with \n if needed.

Key Points:

- Each element in the list is written sequentially.
- Best for writing lines in bulk

